Fiscal Year:	FY 2020	**Task Last Updated:**	FY 08/03/2020
PI Name:	Main, Bob M.S.		
Project Title:	Development of a Goggle-Based Visual Field Device Prototype		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	South Jordan	**State:**	UT
Zip Code:	84009	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	TRISH--Focused Investigations
Start Date:	04/01/2018	**End Date:**	03/31/2020
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 3/31/2020 per E. Urquieta/TRISH (Ed., 5/23/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Buscemi, Philip O.D. (Buscemi Group, LLC)		
Grant/Contract No.:	NNX16AO69A-FIP0006		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Focused Investigation Project Prolonged exposure to microgravity results in multiple alterations to the visual system including hyperopic shifts, globe flattening, papilledema, and choroidal folds. It is essential to monitor the astronauts on a regular basis and to intervene before unresolved significant vision issues arise or permanent vision loss occurs. Currently, the commercial off-the-shelf (COTS) retinal imaging equipment on the International Space Station (ISS) to view/monitor the retina is bulky, requires two astronauts to operate, and is prone to breaking down. As a result, images of the retina are taken very infrequently at a rate of about one time per month and require pupil dilation which incapacitates the astronaut for a few hours. This is currently a high priority need for NASA.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The Goggle-Based Visual Field (GBVF) is a project required by NASA to assess the SANS (Spaceflight Associated Neuro-ocular Syndrome) progression for the astronauts during their missions in space. SANS is a top risk and priority to the International Space Station (ISS) Program Integrated Risk Management Application Risk No. 6169, Human Research Program, HRP TA06, and the Human System Risk Board 3x4 for Deep Space Journey. Currently, several ocular vision tests are performed onboard the ISS, based on NASA Medical Evaluation Documents Volume B 1.10 requirements. One requirement is visual field (VF) testing. VF testing is a well-established, non-invasive test used to assess the effect of ocular and neurological pathology on visual function. Currently, no small-footprint VF device exists to test the impact of SANS on crewmember visual performance during expeditionary spaceflight. The GBVF device is aimed directly at NASA's needs to detect, monitor progression, and guide medical interventions of severe SANS cases during long-duration deep-space missions. It will meet NASA's requirements more quickly and more economically than any other solution.
Task Progress:	During this project (Phase 1), Web Vision Centers Group, LLC (Web Vision) developed a prototype of a novel, ultra-small visual fields device that is goggle shaped/sized for the purposes of conducting visual fields testing of astronauts in-flight. This device (based on NASA's specifications) will detect changes in the astronaut's visual field (VF), monitor visual change progression, and guide medical intervention of Spaceflight Associated Neuro-ocular Syndrome (SANS) cases during long duration space missions. The device is referred to as the Goggle Based Visual Field (GBVF) device. Web Vision has developed a Phase 1 GBVF device (and related software), that is Humphrey compatible/comparable and provides a range of sizes and intensities of stimuli. Note: The Humphrey Perimeter was the first automated visual fields testing device, which reduced variability due to human operators. It has become the de-facto standard to which other perimeters are calibrated and currently it is used in pre and post flight vision testing regime. This device will allow NASA to conduct standard Humphrey 24-2 visual field testing inflight. Currently there is no device on the International Space Station (ISS) to conduct such test because current commercial off-the-shelf (COTS) devices are to large/heavy. Revised Plan - In the original proposal (Phase 1A), we anticipated using a COTS VR (virtual reality) headset and modify it to use for the GBVF prototype. However, after testing many top-quality COTS VR headsets, we discovered that none of them would meet the NASA specifications for the GBVF device. This was communicated to NASA/Translational Research Institute for Space Health (TRISH) in September 2018 and all parties agreed to investigate alternatives. Based on that investigation, we requested a change in the project to allow Web Vision to develop a custom-made headset that enabled the device to meet the NASA specifications. The rescoped project is referred to Phase 1B (May 2019 thru March 2020). The Phase 1B plan was accepted and approved with additional funding. The Phase 1B GBVF device is completed and will be delivered and demonstrated to NASA (and TRISH) in early July 2020. This device is potentially a flight ready device and NASA will assess if it meets all flight specifications.
Bibliography Type:	Description: (Last Updated:)